

SATQUERY AI — REMOTE SENSING ANALYSIS REPORT

ISRO Problem Statement 26167 • Mission ID: b3b40c72 • Generated: 15:34:07.613Z

Natural Query:	Compute NDVI vegetation index and canopy vigor across the spaceport.
Classified Task:	SPECTRAL_INDEX
Calibrated Confidence:	94% (Calibrated via Radiometric Bands)
Specialist Models:	Spectral-Index-Analyzer-v1
Total Latency:	459.7 ms

GROUNDING AI INTERPRETATION

Normalized Difference Vegetation Index (NDVI) was computed across the scene (mean NDVI: -0.0302, range: [-0.5 to 0.3297]). Quantitative analysis reveals 0.0% high-vigor healthy vegetation, 0.21% moderate grassland/shrubland canopy, and 40.99% non-vegetated surfaces (coastal waters, bare ground, and launch complexes). Overall terrain classification: Sparse / Mixed Coastal Terrain.

INPUT SENSOR METADATA

Filename	Modality	Dimensions	Bands	GeoTIFF CRS
isro_sdsc_optical.tif	OPTICAL	512x512	3	EPSG:4326

OBSERVABLE AGENTIC EXECUTION TRACE

#	Phase / Event	Observable Details	Time (m)
1	Query Ingestion	Received natural language query: 'Compute NDVI vegetation index and canopy vigor across the spaceport.' with 1 uploaded image(s).	0.0
2	Input Validation	Verified 1 image file(s). Format and size limits satisfied.	0.0
3	Raster Metadata Extraction: isro_sdsc_optical.tif	Extracted dimensions (512x512), 3 band(s). CRS: EPSG:4326	38.5
4	Modality Identification	Detected sensor modalities: ['OPTICAL'].	0.0
5	Query Classification → SPECTRAL_INDEX	Single image inquiry with spectral / radiometric index mapping intent -> Routed to Spectral Index Specialist (NDVI/NDWI/NDBI).	0.0
6	Model Dispatch	Dispatched task to specialist: 'Spectral-Index-Analyzer-v1' on device 'cpu'.	0.0
7	Specialist Execution Completed	Inference finalized in 421.0ms. Evidence artifacts compiled.	421.0
8	Result Fusion & Calibration	Derived grounded response with calibrated confidence 94% (Calibrated via Radiometric Bands).	0.0